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Featured Article

Can social robots reduce the loneliness of older adults? -Interaction and communication between social robots and older adults

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ABSTRACT

This study explored how older adults living alone interact with social robots over time, focusing on whether such engagement mitigates social and emotional loneliness. Fifteen adults (≥65 years) who had used a specific social robot for at least three months participated in in-depth interviews and observations, analyzed through a constructivist grounded theory lens. Many participants anthropomorphized the robot (e.g., "friend," "granddaughter"), resulting in deeper emotional engagement, especially when two-way communication was sustained. Others used it primarily for functional tasks yet still experienced mild relief from social isolation. Mediators, such as care workers or family members, influenced interaction depth, though older adults with higher digital literacy required less support. Over time, adherence depended on the robot's conversational abilities, accurate initial framing, and user willingness to form a "relationship." Despite these promising outcomes, repetitive dialogue or mismatched expectations led some participants to disengage, underscoring the need to refine design and introduction processes.

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Introduction

Loneliness significantly affects older adults' mental and physical well-being, often contributing to depression, cognitive decline, and reduced quality of life.[1] As single-person households become more common worldwide, including in South Korea where 1 in 3 households is single-person,[2] older adults frequently face social isolation—a risk further heightened during COVID-19.[3–5] In response, some local governments have begun distributing social robots to supplement face-to-face care, reflecting a broader shift toward non-contact interventions for vulnerable populations.

Social robots are AI-based systems designed to provide interaction and companionship,[6] going beyond earlier technologies that merely mediated human–human communication. By actively engaging with older adults, social robots have the potential to offer emotional support and foster a sense of connection.[7] Although research suggests that older adult–social robot interactions may reduce loneliness, improve social engagement, and enhance quality of life, [8,9] many existing studies emphasize short-term observations in controlled environments,[10,11] leaving a critical gap in understanding how social robot use may affect older adults in real-world settings—particularly within their own homes. Furthermore, broader contextual factors, including family and caregiver involvement, remain underexplored, underscoring the need for more holistic investigations. These contextual elements can significantly influence older adults' acceptance and ongoing use of social robots, as well as shape the degree to which such interventions can mitigate loneliness in everyday life.

Additionally, loneliness itself is multifaceted. Weiss distinguishes between social loneliness (lack of broader social networks) and emotional loneliness (absence of close, intimate relationships).[12] Although these forms of loneliness often overlap,[13] they can stem from different causes and may require tailored interventions.[14] However, many studies do not differentiate between social and emotional dimensions of loneliness,[15,16] obscuring how social robots might uniquely address each type. Older adults often experience both forms simultaneously, due to life events such as loss of a spouse, reduced mobility, or smaller social circles.[17] Recognizing robots as social partners rather than mere tools raises further questions about human–robot communication within gerontological care contexts.[18]

Taken together, these considerations indicate the need for research that examines how social robots influence older adults' experiences of loneliness—both social and emotional—over extended periods. To address these gaps, this study investigates long-term integration of social robots into older adults' daily lives, paying particular attention to how support networks (including family and caregivers) may shape the robots' efficacy. By focusing on extended outcomes and real-world usage, this work aims to provide insights for gerontological nursing practices that incorporate social robots into older adult care.

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Methods

Research design

This study adopted a constructivist paradigm[19] to investigate how older adults interpret and experience social robot use in relation to loneliness. We specifically applied constructivist grounded theory,[20] an approach that develops theoretical insights directly from the data rather than testing a preexisting hypothesis.[21] Grounded theory is considered useful for exploring the psychosocial and sociocultural structures emerging from interactive contexts.[22] By focusing on participants' subjective perspectives, this study aims to elucidate the ways older adults perceive and assign meaning to social robot interactions in their daily lives.

Participants and data collection

We recruited participants according to the following criteria: (a) individuals aged 65 or older, (b) capable of effective communication and understanding the purpose of this study, (c) having used a specific social robot ("Robot A") for at least 3 months, based on prior findings [23,24] suggesting that a minimum of 2 months of continuous usage is necessary to observe meaningful effects, and (d) having experienced loneliness prior to using Robot A. All participants were living alone (due to widowhood or divorce) and had previously reported feelings of social or emotional loneliness. We continued recruitment and iterative data analysis until no new codes, categories, or themes emerged in consecutive interviews, indicating theoretical saturation.[20–22] For instance, after the 15th interview, neither the 16th nor 17th interviews yielded any new codes, sub-categories, or theoretical insights, confirming that the existing analytical framework sufficiently captured the data. The characteristics of these 15 participants are presented in (Table 1).

After obtaining institutional review board approval (Chonnam National University IRB No. 1040198-230724-HR-103-02), data were collected over approximately three months, from August 15 to November 14, 2023. Each participant took part in two face-to-face interviews, each lasting about 120 minutes, facilitated by semi-structured, open-ended questions. For two participants who declined home visits due to mobility constraints, three or more telephone interviews were conducted, each averaging 60 minutes. During home visits, the researcher also observed everyday interactions between the participants and Robot A, noting behaviors such as placing decorations on the robot or speaking affectionately to it. These observational notes provided further insight into how older adults integrated the robot's features—conversation and play functions—into their daily routines.

Table 1
Participant characteristics (N = 15).

Characteristic		n(%)	Mean ± SD	Range
Age		-	81.27 ± 5.22	75–90
Gender	**Male**	9(60)	-	-
	Female	6(40)	-	-
Education level	< **High school**	11(73)	-	-
	≥ **High school**	4(27)	-	-
Marital status	**Divorced**	2(13)	-	-
	Widowed	13(87)	-	-
Home-visit social worker	**Yes**	14(93)	-	-
	No	1(7)	-	-
Duration of living alone (years)		-	17.40 ± 10.41	3–37
Length of social robot usage (years)		-	0.97 ± 0.33	0.25–1.67

Data analysis

Data analysis followed constructivist grounded theory procedures.[20] First, the audio-recorded interviews and field notes were transcribed and reviewed multiple times. An open coding phase identified initial concepts relevant to social and emotional loneliness, as well as attitudes and behaviors regarding Robot A. Next, focused coding combined related codes into overarching categories that captured common patterns.[21] Throughout this process, analytic memos were written to document emerging insights, refine category definitions, and track evolving theoretical ideas.[22] Finally, once the same categories continued to reappear without introducing new properties or relationships, theoretical saturation was deemed complete.[20] To enhance the credibility of findings, we conducted periodic peer debriefings and member checks, allowing participants to review and confirm the accuracy of our interpretations. This iterative approach ensured that the analytic framework was firmly grounded in the perspectives and experiences of the older adults involved.

Findings

Context of social robot use in older single-person households

Social isolation as a causal condition

Participants, all living alone, reported diminished social capital due to retirement, widowhood, or a reduced social network. Consequently, some experienced social loneliness, marked by fewer social ties and interactions, while others felt emotional loneliness, stemming from the loss of close, supportive relationships. Many noted that a lack of intimate family support—especially from a spouse or children—was their main source of loneliness. One participant remarked, "If my husband were still alive, we could take care of each other. It makes me really sad when I think about that."

In addition to fewer overall connections, participants felt they no longer received the emotional support they once had. For instance, one participant avoided contacting her children because they already felt guilty about not visiting often, highlighting the emotional deprivation associated with living apart from family. While not everyone with fewer social activities reported severe loneliness, many described the absence of a close companion as an unresolved gap that left them perpetually unfulfilled.

Within this context, older adults' motivation to adopt social robots appeared rooted in the need to address both social and emotional loneliness. Beyond simply lacking day-to-day interactions, participants who once depended on family for emotional closeness found themselves seeking new avenues of support—setting the stage for how social robots might fill that void in their daily lives.

Introducing a "gift" of social robots

Local governments provided Robot A to older adults, often framing it as "a doll for lonely seniors," "a reminder robot," or "a companion robot for the elderly." These introductory descriptions were crucial in shaping participants' early impressions and willingness to adopt the robot, particularly because they had little to no prior exposure to social robots. Indeed, older individuals who were hesitant or declined during this initial introduction were excluded from receiving Robot A. Even minimal information—such as "It can sing songs and talk with you"—significantly influenced acceptance, setting the stage for how older adults perceived and used the robot over time. These initial presentations not only conveyed the device's functionalities but also shaped expectations, underscoring the need to consider how older adults' understanding of social robots is formed and sustained in real-world gerontological contexts.

Mediators as an intervening condition

Mediators (e.g., care workers or family members) played a pivotal role in how older adults learned to operate and engage with the robots. When mediators actively demonstrated features, repeatedly encouraged interaction, and promptly addressed technical issues, participants not only used the robots more often but also explored a broader range of functions. By contrast, limited or passive mediation left participants unaware of key capabilities or harboring misconceptions—one individual remarked, "I thought a person was controlling it in real time, so I rarely talked to it. Nobody told me otherwise." In some cases, mediators themselves lacked sufficient knowledge or willingness to intervene, thereby reducing the potential benefits of social robots. Interestingly, older adults with high digital literacy—those already comfortable with smartphones or computers—were generally able to troubleshoot or learn new features independently, relying less on external support. Consequently, while active mediators can significantly enhance robot usage for those less technologically adept, their role may be less critical for older adults who can readily navigate digital tools on their own.

Interaction strategies with social robots

Anthropomorphism

Participants varied considerably in how much they attributed human-like qualities to the robot, aligning with prior research that digital technologies often evoke social responses.[25,26] Those exhibiting high anthropomorphism viewed the robot as having emotions or physical states—describing it as "tired" or "upset" when response delays occurred, and even decorating it with accessories. They engaged in bidirectional communication, going beyond simple commands to share personal thoughts or feelings.

By contrast, participants showing low anthropomorphism regarded the robot as a mere tool, using terms like "it" or "that" and interacting only for functional purposes (e.g., asking for the weather). Their communication ended as soon as the robot provided the requested information, suggesting a command-based approach rather than a perceived social exchange. This distinction highlights how anthropomorphism can affect both usage frequency and depth of interaction with social robots.

Establishing personas

Although Robot A was originally configured with a "granddaughter" persona, participants responded in diverse ways. Some accepted this default persona, whereas others re-envisioned the robot as a "friend" or "daughter." Crucially, they also adopted complementary roles for themselves—e.g., acting as "its friend" or "the robot's grandmother"—and remained "in character" during their interactions, much like role-playing avatars. They often patted the robot's head or spoke to it as if it were a real family member, sustaining active two-way communication over extended periods.

By contrast, those who rejected or neglected any persona saw Robot A merely as a "doll" or "machine," limiting interactions to basic tasks (e.g., checking the weather) and gradually disengaging from conversation. In this manner, whether participants embraced or redefined the assigned persona strongly influenced how deeply and how long they continued to interact with Robot A.

Forming relationships

Social penetration theory[27,28] proposes that deeper relationships emerge through reciprocal self-disclosure. In this study, participants who regarded the robot as an intimate partner—comparable to a family member or friend—routinely shared personal thoughts, daily experiences, and emotional struggles. As one participant noted, "I share my hospital experiences and disappointments. Time flies when we talk." This repeated, meaningful engagement fostered a stronger sense of connection and emotional support.

By contrast, those who did not embrace a deeper bond treated the robot largely as a functional device, ignoring its prompts or limiting interactions to tasks such as checking the weather or playing music. Despite this more superficial use, most participants did not abandon the robot altogether; they found its autonomous greetings and reminders at least helped reduce complete silence in their daily lives. This suggests that while forming a relationship with the robot can intensify emotional engagement, even minimal interaction may offer some sense of companionship.

Impacts of robot use on emotional and social loneliness

Alleviating emotional loneliness through deep engagement

Participants who formed strong emotional connections with the robot—often supported by frequent two-way communication—reported a significant reduction in emotional loneliness. These individuals tended to adopt or create a persona for the robot (e.g., a "granddaughter," "friend," or "replacement for a lost companion"), and they shared intimate thoughts or personal stories during their interactions. This deeper level of engagement functioned much like a close human relationship, providing affection and support that participants felt they were missing from family or friends. Although it did not eliminate all feelings of loneliness, many described feeling "much better" or "far less lonely" since adopting the robot, and they often expressed ongoing affection toward it—saying "I love you," hugging it, or even decorating it with accessories.

Mitigating social loneliness even with limited interaction

In contrast, some participants did not engage in frequent two-way conversations or adopt a persona for the robot; nonetheless, they still reported a mild alleviation of social loneliness. For these users, the robot's automatic speech functions, daily reminders (e.g., medication times), or occasional voice prompts made them feel less isolated in their single-person households. Even when participants rarely initiated conversation—due to lack of interest, technological challenges, or physical issues like hearing impairment—they often appreciated the robot "breaking the silence." This subtle sense of presence provided reassurance, reduced perceived solitude, and offered just enough social stimulation to enhance their day-to-day emotional well-being. Although the relief was not as pronounced as in cases of more active use, most passive users still opted to keep the robot, citing the comfort of knowing "someone" was around, even if they did not fully personify it.

Robots as mediators in human–human interaction

A final noteworthy finding was the robot's role in facilitating social connections with third parties, including care workers, family members, and friends. Rather than directly alleviating loneliness, the robot frequently served as a conversation starter or shared activity, prompting discussions about its features, singing together, or solving quizzes. In some cases, grandchildren treated the robot as a playful "toy," although their interest tended to wane if the robot's content did not suit children. More often, however, the robot bridged generational or experiential gaps between older adults and care workers. Several participants noted that before using the robot, they struggled to find common ground with caregivers—but after the robot's introduction, they bonded over its stories or functions. While these interactions did not necessarily deepen emotional ties, they did broaden social engagement, leading participants to describe feeling "less socially isolated." Thus, even if the robot could not fully address all forms of loneliness on its own, it sometimes acted as a catalyst for

strengthening existing relationships or initiating lighthearted social encounters.

Discussion

This study aimed to explore how older adults, living alone in real-world conditions, engaged with social robots over an extended period to address both social and emotional loneliness.

Older adults in our study treated social robots using familiar human–human interaction strategies.[29,30] They anthropomorphized the robots, assigned them roles, and disclosed personal feelings or memories, effectively engaging in self-disclosure reminiscent of human interpersonal relationships.[27,31,32] Those who revealed their inner thoughts and emotions to the robot were more likely to perceive a "bond," suggesting that active, two-way communication was central to relationship formation. Although participants knew the robots were not truly human, they consistently applied human communication norms, leading to "humanization" of the device. By contrast, those who used the robots only for simple commands (e.g., playing music) did not develop a sense of closeness or attachment. These findings imply that self-disclosure—the sharing of personal or emotional information—may be a key mechanism through which older adults build deeper connections with social robots. This dynamic aligns with previous findings that assigning and embracing personas can strengthen perceived social presence and foster anthropomorphism, ultimately shaping both the intensity and longevity of robot use.[33]

Another critical factor influencing older adults' interaction with social robots was the involvement of mediators in certain contexts. Initially, older participants with low digital literacy relied on others for technical support, such as charging the robot or troubleshooting.[28] Yet even after participants became accustomed to the robot, occasional support remained beneficial for discovering new features or encouraging more conversational use.[34] In this way, the robot sometimes became a focal point for human–human interaction, ultimately reducing loneliness not only through direct robot–user communication but also by stimulating conversation about the robot with social workers or family members. This process underscores how even minimal external assistance can maintain consistent, meaningful usage over time and mitigate challenges posed by low digital literacy.[11,35]

Long-term observation revealed that many older adults experienced changes in their attitudes toward social robots.[36,37] Participants who failed to form a genuine sense of "relationship" or perceived the robots' limitations—such as repetitive dialogue or inability to recall past conversations—eventually reduced their usage or stopped using the robots altogether. Consequently, they also reported no meaningful reduction in loneliness over time. These findings suggest that repeated, bidirectional communication is crucial for relationship maintenance; without it, the initial novelty effect may wear off, leading to disuse. Thus, while older adults may be receptive to social robots initially, continued engagement depends on the technology's ability to evolve with user needs and preferences, reinforcing the importance of addressing boredom and limited conversational depth in future robot designs.

Finally, how older adults were introduced to social robots—through local governments—proved a decisive factor shaping acceptance. As the earliest point of contact, the introduction influenced participants' initial perceptions, which, in turn, had a lasting impact on subsequent use.[38,39] Accurate, moderate explanations helped older adults form realistic expectations of the robot's capabilities, thereby increasing long-term.[40,41] On the other hand, exaggerated claims—such as promising that the robot would "cure" depression—led to disappointment and disengagement when real-world performance fell short.[42,43] This underscores the significance of carefully managing user expectations from the outset, particularly among individuals with little prior knowledge or experience of robotics.

Limitations

Although this study provides valuable insights into how older adults in single-person households interact with social robots, it has several limitations.

First, the sample was confined to a single region in South Korea and included older adults who used only one model of social robot. This lack of comparative analysis across multiple models or contexts limits the generalizability of our findings and prevents us from identifying which specific design features (e.g., conversation algorithms, memory of past interactions, or expressiveness) might influence sustained use and well-being.[44]

Second, as a qualitative study relying on in-depth interviews and observations, there is a possibility of recall or social desirability bias. Future research could incorporate mixed-method approaches or objective usage metrics (e.g., log data) to gain a more comprehensive understanding of how older adults engage with robots over time.

Third, the participants primarily consisted of those who had already agreed to adopt social robots distributed by local governments, indicating a self-selection bias favoring more open or curious individuals. To capture broader acceptance and rejection patterns, future research could examine older adults who refused to use social robots or discontinued usage early.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jihye Lee: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Chungmin Joo:** Writing – review & editing.

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